

Validation of Change Operations on Workflow Control-Flow Patterns - overview of results for failed cases

This document lists the failure cases from Phase I and the corresponding solution which was achieved using the solution strategies presented in this work. For each failure case, we briefly describe the reason for the failure and present the table with the matrix of the resulting process after implementing the change operation.

WCP-1 (Sequence)

Acceptance sequences: ABC

Insert

- T6: $X \prec_d A; B \prec_d X$ — **Error: Temporal dependencies cause a contradiction.**

	B	C	A	X
B	$\cdot$	$(\prec, \Leftrightarrow)$	$(-, \Leftrightarrow)$	$(\prec_d, \Leftarrow)$
C	$(\succ, \Leftrightarrow)$	$\cdot$	$(\succ, \Leftrightarrow)$	$(\succ, \Leftarrow)$
A	$(-, \Leftrightarrow)$	$(\prec, \Leftrightarrow)$	$\cdot$	$(\succ_d, \Leftarrow)$
X	$(\succ_d, \Rightarrow)$	$(\prec, \Rightarrow)$	$(\prec_d, \Rightarrow)$	$\cdot$

Table 1: Activity relationships matrix for activities B, C, A, X ; using pure occurrence similarity

- E8: $A \Leftrightarrow X; B \not\Leftarrow X$ — **Error: Existential dependencies cause a contradiction.**

	X	A	B	C
X	$\cdot$	$(\prec_d, \Leftrightarrow)$	$\not\Leftarrow$	$(\prec, \Rightarrow)$
A	$(\succ_d, \Leftrightarrow)$	$\cdot$	$\not\Leftarrow$	$(\prec_d, \Rightarrow)$
B	$\not\Leftarrow$	$\not\Leftarrow$	$\cdot$	$(\prec_d, \Rightarrow)$
C	$(\succ, \Leftarrow)$	$(\succ_d, \Leftarrow)$	$(\succ_d, \Leftarrow)$	$\cdot$

Table 2: Activity relationships matrix for activities insert, pure occurrence similarity chosen for skeleton sequence

Modify

- $(A \Leftrightarrow B) \rightarrow (A \Rightarrow B)$ — **Error: does not lead to implication, but equivalence; contradiction between inputs**

	C	B	A
C	$\cdot$	$(\succ_d, \Leftrightarrow)$	$(\succ, \Leftarrow)$
B	$(\prec_d, \Leftrightarrow)$	$\cdot$	$(\succ_d, \Leftarrow)$
A	$(\prec, \Rightarrow)$	$(\prec_d, \Rightarrow)$	$\cdot$

Table 3: Activity relationships matrix for activities C, B, A ; combined occurrence similarity

- $(A \Leftrightarrow B) \rightarrow (A \vee B)$ — **Error: does not lead to OR, but equivalence; contradiction between inputs**

	B	A	C
B	$\cdot$	$(\succ_d, \vee)$	$(\prec_d, \Rightarrow)$
A	$(\prec_d, \vee)$	$\cdot$	$(\prec, \Rightarrow)$
C	$(\succ_d, \Leftarrow)$	$(\succ, \Leftarrow)$	$\cdot$

Table 4: Activity relationships matrix for activities B, A, C ; combined occurrence similarity

- $(A \Leftrightarrow B) \rightarrow (A - B)$ — **Error: does not lead to independence, but equivalence; contradiction between inputs**

	C	A	B
C	$\cdot$	$(\succ, \Leftrightarrow)$	$(\succ_d, \Leftrightarrow)$
A	$(\prec, \Rightarrow)$	$\cdot$	$(\prec_d, -)$
B	$(\prec_d, \Rightarrow)$	$(\succ_d, -)$	$\cdot$

Table 5: Activity relationships matrix for activities C, A, B

- $(A \prec C) \rightarrow (A \prec_d C)$ — **Error: not possible, no final matrix shown; contradiction between inputs**

	B	C	A
B	$\cdot$	$(\succ_d, \Leftrightarrow)$	$(\succ, \Leftrightarrow)$
C	$(\prec_d, \Leftrightarrow)$	$\cdot$	$(\succ_d, \Leftrightarrow)$
A	$(\prec, \Leftrightarrow)$	$(\prec_d, \Leftrightarrow)$	$\cdot$

Table 6: Activity relationships matrix for activities B, C, A ; using mixed occurrence similarity

Collapse

- $\{A, C\}$ — **Error: Activity B happens between the activities to be collapsed.**

	X	B
X	$\cdot$	$(\prec_d, \Leftrightarrow)$
B	$(\succ_d, \Leftrightarrow)$	$\cdot$

Table 7: Activity relationships matrix for moving activities for collapse to activity A

Also possible to expand the set or collapse, only activity X remains

Parallelize

- $\{A, C\}$ — **Error: Activities happening in between.**

	B	A	C
B	$\cdot$	$(-, \Leftrightarrow)$	$(-, \Leftrightarrow)$
A	$(-, \Leftrightarrow)$	$\cdot$	$(-, \Leftrightarrow)$
C	$(-, \Leftrightarrow)$	$(-, \Leftrightarrow)$	$\cdot$

Table 8: Activity relationships matrix for expanding the set of activities to parallelize, including activity B

	C	A	B
C	$\cdot$	$(-, \Leftrightarrow)$	$(\succ, \Leftrightarrow)$
A	$(-, \Leftrightarrow)$	$\cdot$	$(\succ, \Leftrightarrow)$
B	$(\prec, \Leftrightarrow)$	$(\prec, \Leftrightarrow)$	$\cdot$

Table 9: Activity relationships matrix for moving activities for parallelization to activity C

WCP-2 (Parallel Split)

Acceptance sequences: $ABC, ACB, BAC, BCA, CAB, CBA$

Insert

- T6: $X \prec_d A; X \prec_d B$ — **Error: Temporal dependencies cause a contradiction.**

	A	C	X	B
A	$\cdot$	$(-, \Leftrightarrow)$	$(\succ_d, \Leftarrow)$	$(\succ, \Leftrightarrow)$
C	$(-, \Leftrightarrow)$	$\cdot$	$(\prec, \Leftarrow)$	$(-, \Leftrightarrow)$
X	$(\prec_d, \Rightarrow)$	$(\succ, \Rightarrow)$	$\cdot$	$(\succ_d, \Rightarrow)$
B	$(\prec, \Leftrightarrow)$	$(-, \Leftrightarrow)$	$(\prec_d, \Leftarrow)$	$\cdot$

Table 10: Activity relationships matrix for activities A, C, X, B ; occurrence similarity score

- E8: $A \Leftrightarrow X; B \not\Leftarrow X$ — **Error: Existential dependencies cause a contradiction.**

	B	A	X	C
B	$\cdot$	$\not\Leftarrow$	$\not\Leftarrow$	$(\prec_d, \Rightarrow)$
A	$\not\Leftarrow$	$\cdot$	$(\prec, \Leftrightarrow)$	$(-, \Rightarrow)$
X	$\not\Leftarrow$	$(\succ, \Leftrightarrow)$	$\cdot$	$(-, \Rightarrow)$
C	$(\succ_d, \Leftarrow)$	$(-, \Leftarrow)$	$(-, \Leftarrow)$	$\cdot$

Table 11: Activity relationships matrix for activities B, A, X, C ; choosing pure occurrence similarity

Modify

- $(A \Leftrightarrow B) \rightarrow (A \Rightarrow B)$ — **Error: does not lead to implication, but equivalence; contradiction between inputs**

	C	A	B
C	$\cdot$	$(-, \Leftarrow)$	$(-, \Leftrightarrow)$
A	$(-, \Rightarrow)$	$\cdot$	$(-, \Rightarrow)$
B	$(-, \Leftrightarrow)$	$(-, \Leftarrow)$	$\cdot$

Table 12: Activity relationships matrix for activities C, A, B ; combined occurrence similarity

- $(A \Leftrightarrow B) \rightarrow (A \not\Leftarrow B)$ — **Error: not possible, final result not displayed; contradiction between inputs**

	A	C	B
A	$\cdot$	$(-, \Rightarrow)$	$\not\Leftarrow$
C	$(-, \Leftarrow)$	$\cdot$	$(\prec_d, \Leftarrow)$
B	$\not\Leftarrow$	$(\succ_d, \Rightarrow)$	$\cdot$

Table 13: Activity relationships matrix for activities A, C, B

- $(A \Leftrightarrow B) \rightarrow (A \bar{\wedge} B)$ — **Error: not possible, final result not displayed; contradiction between inputs**

	B	C	A
B	$\cdot$	$(\prec_d, \Rightarrow)$	$\overline{\Lambda}$
C	$(\succ_d, \Leftarrow)$	$\cdot$	$(\prec_d, \Leftarrow)$
A	$\overline{\Lambda}$	$(\succ_d, \Rightarrow)$	$\cdot$

Table 14: Activity relationships matrix for activities B, C, A

- $(A \Leftrightarrow B) \rightarrow (A \vee B)$ — **Error: does not lead to OR, but equivalence; contradiction between inputs**

	C	B	A
C	$\cdot$	$(-, \Leftarrow)$	$(-, \Leftarrow)$
B	$(-, \Rightarrow)$	$\cdot$	$(-, \vee)$
A	$(-, \Rightarrow)$	$(-, \vee)$	$\cdot$

Table 15: Activity relationships matrix for activities C, B, A ; combined occurrence similarity

- $(A \Leftrightarrow B) \rightarrow (A - B)$ — **Error: does not lead to independency, but equivalence; contradiction between inputs**

	C	A	B
C	$\cdot$	$(-, \Leftarrow)$	$(-, \Leftarrow)$
A	$(-, \Rightarrow)$	$\cdot$	$(-, -)$
B	$(-, \Rightarrow)$	$(-, -)$	$\cdot$

Table 16: Activity relationships matrix for activities C, A, B

WCP-3 (Synchronization)

Acceptance sequences: ABC, BAC

Insert

- T6: $X \prec_d A; X \prec_d B$ — **Error: Temporal dependencies cause a contradiction.**

	X	B	C	A
X	.	$(\succ_d, \Rightarrow)$	$(\prec, \Rightarrow)$	$(\prec_d, \Rightarrow)$
B	$(\prec_d, \Leftarrow)$	.	$(\prec, \Leftarrow)$	$(\prec, \Leftarrow)$
C	$(\succ, \Leftarrow)$	$(\succ, \Leftarrow)$	.	$(\succ_d, \Leftarrow)$
A	$(\succ_d, \Leftarrow)$	$(\succ, \Leftarrow)$	$(\prec_d, \Leftarrow)$	.

Table 17: Activity relationships matrix for activities X, B, C, A ; occurrence similarity score

- T6: $X \prec_d A; X \prec_d C$ — **Error: Temporal dependencies cause a contradiction.**

	B	C	X	A
B	.	$(\prec, \Leftarrow)$	$(\prec_d, \Leftarrow)$	$(-, \Leftarrow)$
C	$(\succ, \Rightarrow)$	.	$(\succ_d, \vee)$	$(\succ, \vee)$
X	$(\succ_d, \Rightarrow)$	$(\prec_d, \vee)$	.	$(\prec_d, \vee)$
A	$(-, \Rightarrow)$	$(\prec, \vee)$	$(\succ_d, \vee)$	.

Table 18: Activity relationships matrix for activities B, C, X, A ; pure occurrence similarity

- E8: $A \Leftrightarrow X; B \not\Leftarrow X$ — **Error: Existential dependencies cause a contradiction.**

	A	X	B	C
A	.	$(\prec, \Leftarrow)$	$\not\Leftarrow$	$(\prec, \Rightarrow)$
X	$(\succ, \Leftarrow)$	.	$\not\Leftarrow$	$(-, \Rightarrow)$
B	$\not\Leftarrow$	$\not\Leftarrow$	.	$(\prec_d, \Rightarrow)$
C	$(\succ, \Leftarrow)$	$(-, \Leftarrow)$	$(\succ_d, \Leftarrow)$	.

Table 19: Activity relationships matrix for activities A, X, B, C ; choosing pure occurrence similarity

- E8: $A \Leftrightarrow X; C \not\Leftarrow X$ — **Error: Existential dependencies cause a contradiction.**

	B	X	A	C
B	.	$(-, \Leftarrow)$	$(-, \Leftarrow)$	$(\prec_d, \Leftarrow)$
X	$(-, \Rightarrow)$	.	$(\succ, \Leftarrow)$	$\not\Leftarrow$
A	$(-, \Rightarrow)$	$(\prec, \Leftarrow)$	.	$\not\Leftarrow$
C	$(\succ_d, \Rightarrow)$	$\not\Leftarrow$	$\not\Leftarrow$	.

Table 20: Activity relationships matrix for activities B, X, A, C ; pure occurrence similarity

Modify

- $(A \Leftrightarrow C) \rightarrow (A \Rightarrow C)$ — **Error: does not lead to implication, but equivalence; contradiction between inputs**

	C	B	A
C	.	$(\succ, \Leftrightarrow)$	$(\succ, \Leftarrow)$
B	$(\prec, \Leftrightarrow)$	.	$(-, \Leftarrow)$
A	$(\prec, \Rightarrow)$	$(-, \Rightarrow)$	.

Table 21: Activity relationships matrix for activities C, B, A ; combined occurrence similarity

- $(A \Leftrightarrow C) \rightarrow (A \Leftarrow C)$ — **Error: does not lead to implication, but equivalence; contradiction between inputs**

	C	B	A
C	.	$(\succ, \Rightarrow)$	$(\succ, \Rightarrow)$
B	$(\prec, \Leftarrow)$	.	$(-, \Leftrightarrow)$
A	$(\prec, \Leftarrow)$	$(-, \Leftrightarrow)$	.

Table 22: Activity relationships matrix for activities C, B, A ; combined similarity score

- $(A \Leftrightarrow C) \rightarrow (A \not\Leftarrow C)$ — **Error: not possible, does not display a final result; contradiction between inputs**

	C	A	B
C	.	$\not\Leftarrow$	$(\succ_d, \Rightarrow)$
A	$\not\Leftarrow$	.	$(-, \Rightarrow)$
B	$(\prec_d, \Leftarrow)$	$(-, \Leftarrow)$	.

Table 23: Activity relationships matrix for activities C, A, B ; pure occurrence similarity

- $(A \Leftrightarrow C) \rightarrow (A \overline{\Leftarrow} C)$ — **Error: not possible, does not display a final result; contradiction between inputs**

	B	C	A
B	.	$(\prec_d, \Leftarrow)$	$(\prec_d, \Leftarrow)$
C	$(\succ_d, \Rightarrow)$	.	$(-, \overline{\Leftarrow})$
A	$(\succ_d, \Rightarrow)$	$(-, \overline{\Leftarrow})$	.

Table 24: Activity relationships matrix for activities B, C, A ; pure occurrence similarity

- $(A \Leftrightarrow C) \rightarrow (A \vee C)$ — **Error: does not lead to OR, but equivalence; contradiction between inputs**

	B	C	A
B	.	$(\prec, \Leftarrow)$	$(-, \Leftarrow)$
C	$(\succ, \Rightarrow)$	.	$(\succ, \vee)$
A	$(-, \Rightarrow)$	$(\prec, \vee)$	.

Table 25: Activity relationships matrix for activities B, C, A ; combined similarity score

- $(A \Leftrightarrow C) \rightarrow (A - C)$ — **Error: does not lead to independence, but equivalence; contradiction between inputs**

	C	A	B
C	$\cdot$	$(\succ, -)$	$(\succ, \Rightarrow)$
A	$(\prec, -)$	$\cdot$	$(-, \Rightarrow)$
B	$(\prec, \Leftarrow)$	$(-, \Leftarrow)$	$\cdot$

Table 26: Activity relationships matrix for activities C, A, B ; combined similarity score

Collapse

- $\{A, C\}$ — **Error: Activity B happens between the activities to be collapsed.**

	X	B
X	$\cdot$	$(-, \Leftrightarrow)$
B	$(-, \Leftrightarrow)$	$\cdot$

Table 27: Activity relationships matrix for activities X, B ; move all to activity A

Parallelize

- $\{A, C\}$ — **Error: Error, activities happening in between.**

	A	B	C
A	$\cdot$	$(-, \Leftrightarrow)$	$(-, \Leftrightarrow)$
B	$(-, \Leftrightarrow)$	$\cdot$	$(-, \Leftrightarrow)$
C	$(-, \Leftrightarrow)$	$(-, \Leftrightarrow)$	$\cdot$

Table 28: Activity relationships matrix for activities A, B, C ; parallelize by including activity B

WCP-4 (Exclusive Choice)

Acceptance sequences: A, B

Insert

- E8: $A \Leftrightarrow X; B \Leftrightarrow X$ — **Error: Existential dependencies cause a contradiction.**

	A	X	B
A	$\cdot$	$(\prec_d, \Leftrightarrow)$	$(\prec, \Leftrightarrow)$
X	$(\succ_d, \Leftrightarrow)$	$\cdot$	$(\prec_d, \Leftrightarrow)$
B	$(\succ, \Leftrightarrow)$	$(\succ_d, \Leftrightarrow)$	$\cdot$

Table 29: Activity relationships matrix for activities A, X, B ; choosing combined occurrence similarity

Modify

- $(A \not\Leftarrow B) \rightarrow (A \wedge B)$ — **Error: does not lead to NAND, but negated equivalence; contradiction between inputs**

	A	B
A	$\cdot$	$(-, \overline{\wedge})$
B	$(-, \overline{\wedge})$	$\cdot$

Table 30: Activity relationships matrix for activities A, B ; combined occurrence similarity

- $(A \not\Leftarrow B) \rightarrow (A - B)$ — **Error: does not lead to independence, but OR; contradiction between inputs**

	B	A
B	$\cdot$	$(\prec_d, -)$
A	$(\succ_d, -)$	$\cdot$

Table 31: Activity relationships matrix for activities B, A ; combined occurrence similarity

- $(A - B)$, new dependency type $(A \prec B)$ - possible (but does not lead to $\prec$, but independency)

	A	B
A	$\cdot$	$(\prec_d, \vee)$
B	$(\succ_d, \vee)$	$\cdot$

Table 32: Activity relationships matrix for activities A, B ; combined occurrence similarity

Because we do not make a distinction between strict eventual and eventual, this is acceptable

- $(A - B)$, new dependency type $(A \prec_d B)$ - possible (but does not lead to $\prec$, but independency)

	A	B
A	$\cdot$	$(\prec_d, \vee)$
B	$(\succ_d, \vee)$	$\cdot$

Table 33: Activity relationships matrix for activities A, B

WCP-5 (Simple Merge)

Acceptance sequences: AD, BD, CD

Insert

- T6: $X \prec_d A; D \prec_d X$ — **Error: Temporal dependencies cause a contradiction.**

	A	B	C	D	X
A	$\cdot$	$(-, \bar{\Lambda})$	$(-, \bar{\Lambda})$	$(\succ, \Rightarrow)$	$(\succ_d, \Rightarrow)$
B	$(-, \bar{\Lambda})$	$\cdot$	$(-, \bar{\Lambda})$	$(\prec_d, \Rightarrow)$	$(-, \bar{\Lambda})$
C	$(-, \bar{\Lambda})$	$(-, \bar{\Lambda})$	$\cdot$	$(\prec_d, \Rightarrow)$	$(-, \bar{\Lambda})$
D	$(\prec, \Leftarrow)$	$(\succ_d, \Leftarrow)$	$(\succ_d, \Leftarrow)$	$\cdot$	$(\prec_d, \Leftarrow)$
X	$(\prec_d, \Leftarrow)$	$(-, \bar{\Lambda})$	$(-, \bar{\Lambda})$	$(\succ_d, \Rightarrow)$	$\cdot$

Table 34: Activity relationships matrix for activities A, B, C, D, X ; combined similarity score

Modify

- Modify $(A \bar{\Lambda} B) \rightarrow (A - B)$ — **Error: NAND instead of independence; contradiction between inputs**

	B	D	A	C
B	$\cdot$	$(\prec, \Rightarrow)$	$(\prec_d, -)$	$(-, \bar{\Lambda})$
D	$(\succ, \Leftarrow)$	$\cdot$	$(\succ_d, \Leftarrow)$	$(\succ_d, \Leftarrow)$
A	$(\succ_d, -)$	$(\prec_d, \Rightarrow)$	$\cdot$	$(-, \bar{\Lambda})$
C	$(-, \bar{\Lambda})$	$(\prec_d, \Rightarrow)$	$(-, \bar{\Lambda})$	$\cdot$

Table 35: Activity relationships matrix for activities B, D, A, C ; combined occurrence similarity

- Modify $(A \Rightarrow D) \rightarrow (A \bar{\Lambda} D)$ — **Error: negated equivalence instead of NAND; contradiction between inputs**

	D	A	C	B
D	$\cdot$	$(-, \bar{\Lambda})$	$(\succ_d, \Leftarrow)$	$(\succ_d, \Leftarrow)$
A	$(-, \bar{\Lambda})$	$\cdot$	$(-, \bar{\Lambda})$	$(-, \bar{\Lambda})$
C	$(\prec_d, \Rightarrow)$	$(-, \bar{\Lambda})$	$\cdot$	$(-, \bar{\Lambda})$
B	$(\prec_d, \Rightarrow)$	$(-, \bar{\Lambda})$	$(-, \bar{\Lambda})$	$\cdot$

Table 36: Activity relationships matrix for activities D, A, C, B ; combined similarity

- Modify $(A \Rightarrow D) \rightarrow (A - D)$ — **Error: OR instead of independence; contradiction between inputs**

	B	C	D	A
B	$\cdot$	$(-, \bar{\Lambda})$	$(\prec_d, \Rightarrow)$	$(-, \bar{\Lambda})$
C	$(-, \bar{\Lambda})$	$\cdot$	$(\prec_d, \Rightarrow)$	$(-, \bar{\Lambda})$
D	$(\succ_d, \Leftarrow)$	$(\succ_d, \Leftarrow)$	$\cdot$	$(\succ_d, -)$
A	$(-, \bar{\Lambda})$	$(-, \bar{\Lambda})$	$(\prec_d, -)$	$\cdot$

Table 37: Activity relationships matrix for activities B, C, D, A ; combined similarity score

- Modify $(A - B) \rightarrow (A \prec B)$ — **Error: temporal independence instead of $\prec$; contradiction between inputs**

	C	D	A	B
C	$\cdot$	$(\prec_d, \Rightarrow)$	$(-, \overline{\wedge})$	$(-, \overline{\wedge})$
D	$(\succ_d, \Leftarrow)$	$\cdot$	$(\succ, \Leftarrow)$	$(\succ_d, \Leftarrow)$
A	$(-, \overline{\wedge})$	$(\prec, \Rightarrow)$	$\cdot$	$(\prec_d, -)$
B	$(-, \overline{\wedge})$	$(\prec_d, \Rightarrow)$	$(\succ_d, -)$	$\cdot$

Table 38: Activity relationships matrix for activities C, D, A, B ; combined similarity score

Because we do not distinguish between strict eventual and eventual, this is fine.

- Modify $(A - B) \rightarrow (A \prec_d B)$ — **Error: temporal independence instead of $\prec$; contradiction between inputs**

	C	D	A	B
C	$\cdot$	$(\prec_d, \Rightarrow)$	$(-, \overline{\wedge})$	$(-, \overline{\wedge})$
D	$(\succ_d, \Leftarrow)$	$\cdot$	$(\succ, \Leftarrow)$	$(\succ_d, \Leftarrow)$
A	$(-, \overline{\wedge})$	$(\prec, \Rightarrow)$	$\cdot$	$(\prec_d, -)$
B	$(-, \overline{\wedge})$	$(\prec_d, \Rightarrow)$	$(\succ_d, -)$	$\cdot$

Table 39: Activity relationships matrix for activities C, D, A, B ; combined similarity score

WCP-6 (Multi-Choice)

Acceptance sequences: $AB, AC, AD, ABC, \dots, ADCB$

Insert

- T6: $X \prec_d A; B \prec_d X$ — **Error: Temporal dependencies cause a contradiction.**

	A	X	B	D	C
A	$\cdot$	$(\succ_d, \Leftarrow)$	$(-, \Leftarrow)$	$(\prec, \Leftarrow)$	$(\prec, \Leftarrow)$
X	$(\prec_d, \Rightarrow)$	$\cdot$	$(\succ_d, -)$	$(-, \bar{\vee})$	$(-, \bar{\vee})$
B	$(-, \Rightarrow)$	$(\prec_d, -)$	$\cdot$	$(-, -)$	$(-, -)$
D	$(\succ, \Rightarrow)$	$(-, \bar{\vee})$	$(-, -)$	$\cdot$	$(-, -)$
C	$(\succ, \Rightarrow)$	$(-, \bar{\vee})$	$(-, -)$	$(-, -)$	$\cdot$

Table 40: Activity relationships matrix for activities A, X, B, D, C ; combined similarity

- T6: $X \prec_d C; X \prec_d B$ — **Error: Temporal dependencies cause a contradiction.**

	B	C	D	A	X
B	$\cdot$	$(-, -)$	$(-, -)$	$(\succ, \Rightarrow)$	$(\succ_d, -)$
C	$(-, -)$	$\cdot$	$(-, -)$	$(\succ, \Rightarrow)$	$(\succ_d, -)$
D	$(-, -)$	$(-, -)$	$\cdot$	$(\succ, \Rightarrow)$	$(-, \bar{\vee})$
A	$(\prec, \Leftarrow)$	$(\prec, \Leftarrow)$	$(\prec, \Leftarrow)$	$\cdot$	$(\prec_d, \Leftarrow)$
X	$(\prec_d, -)$	$(\prec_d, -)$	$(-, \bar{\vee})$	$(\succ_d, \Rightarrow)$	$\cdot$

Table 41: Activity relationships matrix for activities B, C, D, A, X ; combined sim score

- E8: $A \Leftrightarrow X; B \not\Leftarrow X$ — **Error: Activity B not part of the process; contradiction between inputs**

	C	B	D	A	X
C	$\cdot$	$(-, \bar{\vee})$	$(-, -)$	$(\succ, \Rightarrow)$	$(-, \Rightarrow)$
B	$(-, \bar{\vee})$	$\cdot$	$(-, \bar{\vee})$	$(-, \not\Leftarrow)$	$(-, \not\Leftarrow)$
D	$(-, -)$	$(-, \bar{\vee})$	$\cdot$	$(\succ, \Rightarrow)$	$(-, \Rightarrow)$
A	$(\prec, \Leftarrow)$	$(-, \not\Leftarrow)$	$(\prec, \Leftarrow)$	$\cdot$	$(\prec, \Leftarrow)$
X	$(-, \Leftarrow)$	$(-, \not\Leftarrow)$	$(-, \Leftarrow)$	$(\succ, \Leftarrow)$	$\cdot$

Table 42: Activity relationships matrix for activities C, B, D, A, X ; combined similarity score

Modify

- Modify $(A \Rightarrow B) \rightarrow (A \bar{\vee} B)$ — **Error: instead of NAND we get negated equivalence; contradiction between inputs**

	D	A	B	C
D	$\cdot$	$(\succ, \Rightarrow)$	$(-, \bar{\vee})$	$(-, -)$
A	$(\prec, \Leftarrow)$	$\cdot$	$(-, \bar{\vee})$	$(\prec, \Leftarrow)$
B	$(-, \bar{\vee})$	$(-, \bar{\vee})$	$\cdot$	$(-, \bar{\vee})$
C	$(-, -)$	$(\succ, \Rightarrow)$	$(-, \bar{\vee})$	$\cdot$

Table 43: Activity relationships matrix for activities D, A, B, C ; combined similarity score

- Modify $(A \Rightarrow B) \rightarrow (A - B)$ — **Error: instead of independence we get OR; contradiction between inputs**

	B	C	D	A
B	$\cdot$	$(-, -)$	$(-, -)$	$(\succ, -)$
C	$(-, -)$	$\cdot$	$(-, -)$	$(\succ, \Rightarrow)$
D	$(-, -)$	$(-, -)$	$\cdot$	$(\succ, \Rightarrow)$
A	$(\prec, -)$	$(\prec, \Leftarrow)$	$(\prec, \Leftarrow)$	$\cdot$

Table 44: Activity relationships matrix for activities B, C, D, A

- Modify $(B-C) \rightarrow (B \Rightarrow C)$ — **Error: instead of implication we get equivalence; contradiction between inputs**

	A	D	C	B
A	$\cdot$	$(\prec, \Leftarrow)$	$(\prec, \Leftarrow)$	$(\prec, \Leftarrow)$
D	$(\succ, \Rightarrow)$	$\cdot$	$(-, -)$	$(-, -)$
C	$(\succ, \Rightarrow)$	$(-, -)$	$\cdot$	$(-, \Leftarrow)$
B	$(\succ, \Rightarrow)$	$(-, -)$	$(-, \Rightarrow)$	$\cdot$

Table 45: Activity relationships matrix for activities A, D, C, B

Collapse

- $\{A, B\}$ — **Error: Activities happening in between.**

	X	C	D
X	$\cdot$	$(\prec, \Leftarrow)$	$(\prec, \Leftarrow)$
C	$(\succ, \Rightarrow)$	$\cdot$	$(-, -)$
D	$(\succ, \Rightarrow)$	$(-, -)$	$\cdot$

Table 46: Activity relationships matrix for activities X, C, D ; collapse by moving all activities to activity B

Parallelize

- $\{A, B\}$ — **Error: Activities happening in between.**

	C	A	B	D
C	$\cdot$	$(-, \Leftrightarrow)$	$(-, \Leftrightarrow)$	$(-, \Leftrightarrow)$
A	$(-, \Leftrightarrow)$	$\cdot$	$(-, \Leftrightarrow)$	$(-, \Leftrightarrow)$
B	$(-, \Leftrightarrow)$	$(-, \Leftrightarrow)$	$\cdot$	$(-, \Leftrightarrow)$
D	$(-, \Leftrightarrow)$	$(-, \Leftrightarrow)$	$(-, \Leftrightarrow)$	$\cdot$

Table 47: Activity relationships matrix for activities C, A, B, D ; Parallelize by including activities

	B	D	C	A
B	$\cdot$	$(-, -)$	$(-, -)$	$(-, \Leftrightarrow)$
D	$(-, -)$	$\cdot$	$(-, -)$	$(-, -)$
C	$(-, -)$	$(-, -)$	$\cdot$	$(-, -)$
A	$(-, \Leftrightarrow)$	$(-, -)$	$(-, -)$	$\cdot$

Table 48: Activity relationships matrix for activities B, D, C, A ; parallelize by moving all activities to activity B

WCP-7 (Structured Synchronizing Merge)

Acceptance sequences: AD , $BD, \dots, CBAD$

Insert

- T6: $X \prec_d A$; $D \prec_d X$ — **Error: Temporal dependencies cause a contradiction.**

	D	A	B	X	C
D	$\cdot$	$(-, \Leftarrow)$	$(\succ, \Leftarrow)$	$(\prec_d, \Leftarrow)$	$(\succ, \Leftarrow)$
A	$(-, \Rightarrow)$	$\cdot$	$(-, -)$	$(\succ_d, -)$	$(-, -)$
B	$(\prec, \Rightarrow)$	$(-, -)$	$\cdot$	$(-, \overline{\wedge})$	$(-, -)$
X	$(\succ_d, \Rightarrow)$	$(\prec_d, -)$	$(-, \overline{\wedge})$	$\cdot$	$(-, \overline{\wedge})$
C	$(\prec, \Rightarrow)$	$(-, -)$	$(-, -)$	$(-, \overline{\wedge})$	$\cdot$

Table 49: Activity relationships matrix for activities D, A, B, X, C

Collapse

- $\{A, B, D\}$ — **Error: Activity C happens between the activities to be collapsed.**

	C	X
C	$\cdot$	$(\prec_d, \Rightarrow)$
X	$(\succ_d, \Leftarrow)$	$\cdot$

Table 50: Activity relationships matrix for activities C, X ; collapse by moving all activities to activity D

Parallelize

- $\{A, B, D\}$ — **Error: Error, activities happening in between.**

	C	D	B	A
C	$\cdot$	$(-, -)$	$(-, -)$	$(-, -)$
D	$(-, -)$	$\cdot$	$(-, \Leftrightarrow)$	$(-, \Leftrightarrow)$
B	$(-, -)$	$(-, \Leftrightarrow)$	$\cdot$	$(-, \Leftrightarrow)$
A	$(-, -)$	$(-, \Leftrightarrow)$	$(-, \Leftrightarrow)$	$\cdot$

Table 51: Activity relationships matrix for activities C, D, B, A ; parallelize by moving all to activity A

WCP-10 (Arbitrary Cycles)

Acceptance sequences: $ABC, ABCBC, \dots, C, BC,$

...

Insert

- T6: $X \prec_d A; B \prec X$ — **Error: Temporal dependencies cause a contradiction.**

	X	A	C	B
X	$\cdot$	$(\prec_d, -)$	$(-, \Rightarrow)$	$(\succ, \Rightarrow)$
A	$(\succ_d, -)$	$\cdot$	$(-, \Rightarrow)$	$(-, \Rightarrow)$
C	$(-, \Leftarrow)$	$(-, \Leftarrow)$	$\cdot$	$(\succ, \Leftarrow)$
B	$(\prec, \Leftarrow)$	$(-, \Leftarrow)$	$(\prec, \Rightarrow)$	$\cdot$

Table 52: Activity relationships matrix for activities X, A, C, B ; combined occurrence similarity

Modify

- Modify $(B \Rightarrow C)$, new dependency type $(B \nRightarrow C)$ - possible (A no longer part of the process (contradiction))

	A	C	B
A	$\cdot$	$(-, \overline{\wedge})$	$(\prec_d, \Rightarrow)$
C	$(-, \overline{\wedge})$	$\cdot$	$(-, \nRightarrow)$
B	$(\succ_d, \Leftarrow)$	$(-, \nRightarrow)$	$\cdot$

Table 53: Activity relationships matrix for activities A, C, B ; combined similarity score

- Modify $(A \Rightarrow C)$, new dependency type $(A \nRightarrow C)$ - possible (A no longer in the matrix (contradiction))

	B	C	A
B	$\cdot$	$(\prec_d, \vee)$	$(\succ_d, \Leftarrow)$
C	$(\succ_d, \vee)$	$\cdot$	$(-, \nRightarrow)$
A	$(\prec_d, \Rightarrow)$	$(-, \nRightarrow)$	$\cdot$

Table 54: Activity relationships matrix for activities B, C, A

- Modify $(B \Rightarrow C)$, new dependency type $(B \overline{\wedge} C)$ - possible (negated existential equivalence instead of NAND, A no longer part of the process)

	A	C	B
A	$\cdot$	$(-, \overline{\wedge})$	$(-, \overline{\wedge})$
C	$(-, \overline{\wedge})$	$\cdot$	$(-, \overline{\wedge})$
B	$(-, \overline{\wedge})$	$(-, \overline{\wedge})$	$\cdot$

Table 55: Activity relationships matrix for activities A, C, B

- Modify $(A \prec_d B)$, new dependency type $(A \succ B)$ - possible (A no longer part of the process (contradiction))

	A	C	B
A	$\cdot$	$(-, \Rightarrow)$	$(\succ, \Rightarrow)$
C	$(-, \Leftarrow)$	$\cdot$	$(\succ, \Leftarrow)$
B	$(\prec, \Leftarrow)$	$(\prec, \Rightarrow)$	$\cdot$

Table 56: Activity relationships matrix for activities A, C, B ; combined similarity score

- Modify $(A \prec C)$, new dependency type $(A \prec_d C)$ - possible (A no longer part of the process (contradiction))

	A	C	B
A	$\cdot$	$(\prec_d, \Rightarrow)$	$(\prec, \Rightarrow)$
C	$(\succ_d, \Leftarrow)$	$\cdot$	$(-, \Leftarrow)$
B	$(\succ, \Leftarrow)$	$(-, \Rightarrow)$	$\cdot$

Table 57: Activity relationships matrix for activities A, C, B

- Modify $(A \prec C)$, new dependency type $(A \succ C)$ - possible (A no longer part of the process (contradiction))

	B	A	C
B	$\cdot$	$(-, \overline{\wedge})$	$(\prec_d, \vee)$
A	$(-, \overline{\wedge})$	$\cdot$	$(\succ_d, \Rightarrow)$
C	$(\succ_d, \vee)$	$(\prec_d, \Leftarrow)$	$\cdot$

Table 58: Activity relationships matrix for activities B, A, C ; combined similarity score

- Modify $(A \prec C) \rightarrow (A-C)$ — **Error: eventual temporal dependency instead of independence; contradiction between inputs**

	B	C	A
B	$\cdot$	$(\prec_d, \Rightarrow)$	$(\succ_d, -)$
C	$(\succ_d, \Leftarrow)$	$\cdot$	$(-, \Leftarrow)$
A	$(\prec_d, -)$	$(-, \Rightarrow)$	$\cdot$

Table 59: Activity relationships matrix for activities B, C, A ; combined similarity score

- Modify $(B \prec_d C)$, new dependency type $(B \succ C)$ - possible (A no longer part of the process (contradiction))

	A	C	B
A	$\cdot$	$(-, \overline{\wedge})$	$(-, \overline{\wedge})$
C	$(-, \overline{\wedge})$	$\cdot$	$(\prec_d, \Leftarrow)$
B	$(-, \overline{\wedge})$	$(\succ_d, \Rightarrow)$	$\cdot$

Table 60: Activity relationships matrix for activities A, C, B

Collapse

- $\{A, C\}$ — **Error: Activity B happens between the activities to be collapsed.**

	B	X
B	$\cdot$	$(\succ_d, \Leftarrow)$
X	$(\prec_d, \Rightarrow)$	$\cdot$

Table 61: Activity relationships matrix for activities B , X ; move all to activity A

Parallelize

- $\{A, C\}$ — **Error: Activity B happens between the activities to be parallelized.**

	A	C	B
A	$\cdot$	$(-, \Leftrightarrow)$	$(\succ, \Leftarrow)$
C	$(-, \Leftrightarrow)$	$\cdot$	$(\succ, \Leftarrow)$
B	$(\prec, \Rightarrow)$	$(\prec, \Rightarrow)$	$\cdot$

Table 62: Activity relationships matrix for activities A , C , B ; move all to activity C

WCP-17 (Interleaved Parallel Routing)

Acceptance sequences: ABC , ACB , CAB

Insert

- T6: $X \prec_d A$; $B \prec X$ — **Error: Temporal dependencies cause a contradiction.**

	A	C	B	X
A	.	$(-, \Leftrightarrow)$	$(\succ, \Leftrightarrow)$	$(\succ_d, \Leftarrow)$
C	$(-, \Leftrightarrow)$	.	$(-, \Leftrightarrow)$	$(\prec_d, \Leftarrow)$
B	$(\prec, \Leftrightarrow)$	$(-, \Leftrightarrow)$	.	$(\prec, \Leftarrow)$
X	$(\prec_d, \Rightarrow)$	$(\succ_d, \Rightarrow)$	$(\succ, \Rightarrow)$	.

Table 63: Activity relationships matrix for activities A, C, B, X ; occurrence similarity score

- E8: $A \Leftrightarrow X$; $C \not\Leftarrow X$ — **Error: Existential dependencies cause a contradiction.**

	B	C	X	A
B	.	$(\succ_d, \Leftarrow)$	$(-, \Leftarrow)$	$(\succ, \Leftarrow)$
C	$(\prec_d, \Rightarrow)$	.	$\not\Leftarrow$	$\not\Leftarrow$
X	$(-, \Rightarrow)$	$\not\Leftarrow$	.	$(\succ, \Leftrightarrow)$
A	$(\prec, \Rightarrow)$	$\not\Leftarrow$	$(\prec, \Leftrightarrow)$	.

Table 64: Activity relationships matrix for activities B, C, X, A ; pure occurrence similarity

Modify

- $(A \Leftrightarrow B) \rightarrow (A \not\Leftarrow B)$ — **Error: not possible, contradiction between inputs.**

	B	A	C
B	.	$(-, \not\Leftarrow)$	$(\prec_d, \Rightarrow)$
A	$(-, \not\Leftarrow)$	.	$(-, \Rightarrow)$
C	$(\succ_d, \Leftarrow)$	$(-, \Leftarrow)$	.

Table 65: Activity relationships matrix for activities B, A, C ; combined sim score

- $(A \Leftrightarrow B) \rightarrow (A \bar{\wedge} B)$ — **Error: not possible, contradiction between inputs.**

	C	A	B
C	.	$(\prec_d, \Leftarrow)$	$(\prec_d, \Leftarrow)$
A	$(\succ_d, \Rightarrow)$	.	$(-, \bar{\wedge})$
B	$(\succ_d, \Rightarrow)$	$(-, \bar{\wedge})$	.

Table 66: Activity relationships matrix for activities C, A, B ; combined sim score

- $(A \Leftrightarrow C) \rightarrow (A \not\Leftarrow C)$ — **Error: not possible, contradiction between inputs.**

	A	B	C
A	.	$(\prec_d, \Rightarrow)$	$(-, \not\Leftarrow)$
B	$(\succ_d, \Leftarrow)$	.	$(\prec_d, \Leftarrow)$
C	$(-, \not\Leftarrow)$	$(\succ_d, \Rightarrow)$	.

Table 67: Activity relationships matrix for activities A, B, C ; combined sim score

- $(A \Leftrightarrow C) \rightarrow (A \overline{\wedge} C)$ — **Error: not possible, contradiction between inputs.**

	A	B	C
A	.	$(\prec_d, \Rightarrow)$	$(-, \overline{\wedge})$
B	$(\succ_d, \Leftarrow)$	.	$(\prec_d, \Leftarrow)$
C	$(-, \overline{\wedge})$	$(\succ_d, \Rightarrow)$	.

Table 68: Activity relationships matrix for activities A, B, C ; combined sim score

- Modify $(A \Leftrightarrow B) \rightarrow (A \Rightarrow B)$ — **Error: existential equivalence instead of implication; contradiction between inputs**

	C	A	B
C	.	$(-, \Leftarrow)$	$(-, \Leftrightarrow)$
A	$(-, \Rightarrow)$	.	$(\prec, \Rightarrow)$
B	$(-, \Leftrightarrow)$	$(\succ, \Leftarrow)$	.

Table 69: Activity relationships matrix for activities C, A, B ; combined similarity score

- Modify $(A \Leftrightarrow B) \rightarrow (A \Leftarrow B)$ — **Error: existential equivalence instead of implication; contradiction between inputs**

	B	C	A
B	.	$(-, \Rightarrow)$	$(\succ, \Rightarrow)$
C	$(-, \Leftarrow)$	.	$(-, \Leftrightarrow)$
A	$(\prec, \Leftarrow)$	$(-, \Leftrightarrow)$	.

Table 70: Activity relationships matrix for activities B, C, A ; combined similarity score

- Modify $(A \Leftrightarrow B) \rightarrow (A \vee B)$ — **Error: existential equivalence instead of OR; contradiction between inputs**

	B	A	C
B	.	$(\succ, \vee)$	$(-, \Rightarrow)$
A	$(\prec, \vee)$	.	$(-, \Rightarrow)$
C	$(-, \Leftarrow)$	$(-, \Leftarrow)$	.

Table 71: Activity relationships matrix for activities B, A, C ; combined similarity score

- Modify $(A \Leftrightarrow B) \rightarrow (A - B)$ — **Error: existential equivalence instead of independence; contradiction between inputs**

	A	C	B
A	.	$(-, \Rightarrow)$	$(\prec, -)$
C	$(-, \Leftarrow)$	.	$(-, \Leftarrow)$
B	$(\succ, -)$	$(-, \Rightarrow)$	.

Table 72: Activity relationships matrix for activities A, C, B ; combined similarity score

- Modify $(A \Leftrightarrow C) \rightarrow (A \Rightarrow C)$ — **Error: existential equivalence instead of implication; contradiction between inputs**

	A	C	B
A	$\cdot$	$(-, \Rightarrow)$	$(\prec, \Rightarrow)$
C	$(-, \Leftarrow)$	$\cdot$	$(-, \Leftrightarrow)$
B	$(\succ, \Leftarrow)$	$(-, \Leftrightarrow)$	$\cdot$

Table 73: Activity relationships matrix for activities A, C, B ; combined similarity score

- Modify $(A \Leftrightarrow C) \rightarrow (A \Leftarrow C)$ — **Error: existential equivalence instead of implication; contradiction between inputs**

	C	A	B
C	$\cdot$	$(-, \Rightarrow)$	$(-, \Rightarrow)$
A	$(-, \Leftarrow)$	$\cdot$	$(\prec, \Leftrightarrow)$
B	$(-, \Leftarrow)$	$(\succ, \Leftrightarrow)$	$\cdot$

Table 74: Activity relationships matrix for activities C, A, B ; combined similarity score

- Modify $(A \Leftrightarrow C) \rightarrow (A \vee C)$ — **Error: existential equivalence instead of OR; contradiction between inputs**

	B	C	A
B	$\cdot$	$(-, \Leftarrow)$	$(\succ, \Leftarrow)$
C	$(-, \Rightarrow)$	$\cdot$	$(-, \vee)$
A	$(\prec, \Rightarrow)$	$(-, \vee)$	$\cdot$

Table 75: Activity relationships matrix for activities B, C, A ; combined similarity score

- Modify $(A \Leftrightarrow C) \rightarrow (A - C)$ — **Error: existential equivalence instead of independence; contradiction between inputs**

	B	C	A
B	$\cdot$	$(-, \Leftarrow)$	$(\succ, \Leftarrow)$
C	$(-, \Rightarrow)$	$\cdot$	$(-, -)$
A	$(\prec, \Rightarrow)$	$(-, -)$	$\cdot$

Table 76: Activity relationships matrix for activities B, C, A ; combined similarity score

Collapse

- $\{A, C\}$ — **Error: Activity B happens between the activities to be collapsed.**

	X	B
X	$\cdot$	$(-, \Leftrightarrow)$
B	$(-, \Leftrightarrow)$	$\cdot$

Table 77: Activity relationships matrix for activities X, B ; move to activity C to collapse

- $\{B, C\}$ — **Error: Activity A happens between the activities to be collapsed.**

	A	X
A	$\cdot$	$(-, \Leftrightarrow)$
X	$(-, \Leftrightarrow)$	$\cdot$

Table 78: Activity relationships matrix for activities A, X ; move to activity C to collapse

Parallelize

- $\{A, C\}$ — **Error: Error, activities happening between**

	A	B	C
A	$\cdot$	$(\prec, \Leftrightarrow)$	$(-, \Leftrightarrow)$
B	$(\succ, \Leftrightarrow)$	$\cdot$	$(\succ, \Leftrightarrow)$
C	$(-, \Leftrightarrow)$	$(\prec, \Leftrightarrow)$	$\cdot$

Table 79: Activity relationships matrix for activities A, B, C ; move all to activity A to parallelize

- $\{B, C\}$ — **Error: Error, activities happening between**

	C	A	B
C	$\cdot$	$(-, \Leftrightarrow)$	$(-, \Leftrightarrow)$
A	$(-, \Leftrightarrow)$	$\cdot$	$(-, \Leftrightarrow)$
B	$(-, \Leftrightarrow)$	$(-, \Leftrightarrow)$	$\cdot$

Table 80: Activity relationships matrix for activities C, A, B ; move all to activity C to parallelize

WCP-19 (Cancel Activity)

Acceptance sequences: AB, A

Insert

- T6: $A \prec_d X; B \prec_d X$ — **Error: Temporal dependencies cause a contradiction.**

	A	B	X
A	$\cdot$	$(-, \Leftarrow)$	$(\succ_d, \Leftarrow)$
B	$(-, \Rightarrow)$	$\cdot$	$(\prec_d, -)$
X	$(\prec_d, \Rightarrow)$	$(\succ_d, -)$	$\cdot$

Table 81: Activity relationships matrix for activities A, B, X ; combined sim score

Locked Dependencies

Sequential Process ($A \rightarrow B \rightarrow C \rightarrow D$)

Locked dependency $A \prec_d B$

- Swap A and B — **Error: Temporal dependency from 'A' to 'B' is locked and cannot be changed.**

	C	B	A	D
C	.	$(\succ, \Leftrightarrow)$	$(\succ_d, \Leftrightarrow)$	$(\prec_d, \Leftrightarrow)$
B	$(\prec, \Leftrightarrow)$	.	$(\prec_d, \Leftrightarrow)$	$(\prec, \Leftrightarrow)$
A	$(\prec_d, \Leftrightarrow)$	$(\succ_d, \Leftrightarrow)$	.	$(\prec, \Leftrightarrow)$
D	$(\succ_d, \Leftrightarrow)$	$(\succ, \Leftrightarrow)$	$(\succ, \Leftrightarrow)$	.

Table 82: Activity relationships matrix for activities C, B, A, D ; possible by deleting the locked dependency

- Swap A and C — **Error: Temporal dependency from 'A' to 'B' is locked and cannot be changed.**

	A	C	D	B
A	.	$(\succ_d, \Leftrightarrow)$	$(\prec, \Leftrightarrow)$	$(\prec_d, \Leftrightarrow)$
C	$(\prec_d, \Leftrightarrow)$	.	$(\prec, \Leftrightarrow)$	$(\prec, \Leftrightarrow)$
D	$(\succ, \Leftrightarrow)$	$(\succ, \Leftrightarrow)$	.	$(\succ_d, \Leftrightarrow)$
B	$(\succ_d, \Leftrightarrow)$	$(\succ, \Leftrightarrow)$	$(\prec_d, \Leftrightarrow)$	.

Table 83: Activity relationships matrix for activities A, C, D, B ; pure occurrence similarity

- Parallelize $\{A, B\}$ — **Error: Temporal dependency from 'A' to 'B' is locked and cannot be changed.**

	B	A	C	D
B	.	$(-, \Leftrightarrow)$	$(\prec, \Leftrightarrow)$	$(\prec, \Leftrightarrow)$
A	$(-, \Leftrightarrow)$	.	$(\prec, \Leftrightarrow)$	$(\prec, \Leftrightarrow)$
C	$(\succ, \Leftrightarrow)$	$(\succ, \Leftrightarrow)$	.	$(\prec_d, \Leftrightarrow)$
D	$(\succ, \Leftrightarrow)$	$(\succ, \Leftrightarrow)$	$(\succ_d, \Leftrightarrow)$	.

Table 84: Activity relationships matrix for activities B, A, C, D ; possible by deleting locked dependency

- Parallelize $\{B, C\}$ — **Error: Temporal dependency from 'A' to 'B' is locked and cannot be changed.**

	D	C	A	B
D	.	$(\succ, \Leftrightarrow)$	$(\succ, \Leftrightarrow)$	$(\succ, \Leftrightarrow)$
C	$(\prec, \Leftrightarrow)$	.	$(\succ, \Leftrightarrow)$	$(-, \Leftrightarrow)$
A	$(\prec, \Leftrightarrow)$	$(\prec, \Leftrightarrow)$	.	$(\prec, \Leftrightarrow)$
B	$(\prec, \Leftrightarrow)$	$(-, \Leftrightarrow)$	$(\succ, \Leftrightarrow)$	.

Table 85: Activity relationships matrix for activities D, C, A, B ; using dependency relaxation to perform the change operation

	C	A	B	D
C	.	$(-, \Leftrightarrow)$	$(-, \Leftrightarrow)$	$(\prec, \Leftrightarrow)$
A	$(-, \Leftrightarrow)$	.	$(\prec_d, \Leftrightarrow)$	$(\prec, \Leftrightarrow)$
B	$(-, \Leftrightarrow)$	$(\succ_d, \Leftrightarrow)$	.	$(\prec, \Leftrightarrow)$
D	$(\succ, \Leftrightarrow)$	$(\succ, \Leftrightarrow)$	$(\succ, \Leftrightarrow)$	.

Table 86: Activity relationships matrix for activities C, A, B, D ; no dependency relaxation but combined occurrence similarity

Locked dependency $A \Leftrightarrow B$

- Delete A — **Error: Activity 'A' cannot be deleted because existential dependency from 'A' to 'B' is locked.**

	D	B	C
D	.	$(\succ, \Leftrightarrow)$	$(\succ_d, \Leftrightarrow)$
B	$(\prec, \Leftrightarrow)$	.	$(\prec_d, \Leftrightarrow)$
C	$(\prec_d, \Leftrightarrow)$	$(\succ_d, \Leftrightarrow)$	.

Table 87: Activity relationships matrix for activities D, B, C ; deleting the locked dependency in dialogue

- Collapse $\{A, B\}$ — **Error: Activity 'A' cannot be deleted because existential dependency from 'A' to 'B' is locked.**

	C	D	X
C	.	$(\prec_d, \Leftrightarrow)$	$(\succ_d, \Leftrightarrow)$
D	$(\succ_d, \Leftrightarrow)$	.	$(\succ, \Leftrightarrow)$
X	$(\prec_d, \Leftrightarrow)$	$(\prec, \Leftrightarrow)$	.

Table 88: Activity relationships matrix for activities C, D, X ; dependencies are only part of the collapsed set; so they get removed

- Collapse $\{B, C\}$ — **Error: Activity 'B' cannot be deleted because existential dependency from 'A' to 'B' is locked.**

	D	A	X
D	.	$(\succ, \Leftrightarrow)$	$(\succ_d, \Leftrightarrow)$
A	$(\prec, \Leftrightarrow)$	.	$(\prec_d, \Leftrightarrow)$
X	$(\prec_d, \Leftrightarrow)$	$(\succ_d, \Leftrightarrow)$	.

Table 89: Activity relationships matrix for activities D, A, X ; delete the locked dependency in dialogue with user

- De-collapse A — **Error: Activity 'A' cannot be deleted because existential dependency from 'A' to 'B' is locked.**

	C	Y	X	D	B
C	.	$(\succ, \Leftarrow)$	$(\succ, \Leftarrow)$	$(\prec_d, \Leftrightarrow)$	$(\succ_d, \Leftrightarrow)$
Y	$(\prec, \Rightarrow)$	.	$(\succ_d, \Rightarrow)$	$(\prec, \Rightarrow)$	$(\prec_d, \Rightarrow)$
X	$(\prec, \Rightarrow)$	$(\prec_d, \Leftarrow)$	.	$(\prec, \Rightarrow)$	$(\prec, \Rightarrow)$
D	$(\succ_d, \Leftrightarrow)$	$(\succ, \Leftarrow)$	$(\succ, \Leftarrow)$	.	$(\succ, \Leftarrow)$
B	$(\prec_d, \Leftrightarrow)$	$(\succ_d, \Leftarrow)$	$(\succ, \Leftarrow)$	$(\prec, \Leftrightarrow)$	.

Table 90: Activity relationships matrix for activities C, Y, X, D, B ; delete locked dependency in dialogue with the user

- Replace A — **Error: Activity 'A' cannot be deleted because existential dependency from 'A' to 'B' is locked.**

	C	D	X	B
C	.	$(\prec_d, \Leftrightarrow)$	$(\succ, \Leftrightarrow)$	$(\succ_d, \Leftrightarrow)$
D	$(\succ_d, \Leftrightarrow)$	.	$(\succ, \Leftrightarrow)$	$(\succ, \Leftrightarrow)$
X	$(\prec, \Leftrightarrow)$	$(\prec, \Leftrightarrow)$	.	$(\prec_d, \Leftrightarrow)$
B	$(\prec_d, \Leftrightarrow)$	$(\prec, \Leftrightarrow)$	$(\succ_d, \Leftrightarrow)$	.

Table 91: Activity relationships matrix for activities C, D, X, B ; delete locked dependency in dialogue with the user

- Skip A — **Error: Activity 'A' cannot be deleted because existential dependency from 'A' to 'B' is locked.**

	B	A	C	D
B	.	$(\succ_d, \Leftarrow)$	$(\prec_d, \Leftrightarrow)$	$(\prec, \Leftrightarrow)$
A	$(\prec_d, \Rightarrow)$	.	$(\prec, \Rightarrow)$	$(\prec, \Rightarrow)$
C	$(\succ_d, \Leftrightarrow)$	$(\succ, \Leftarrow)$	.	$(\prec_d, \Leftrightarrow)$
D	$(\succ, \Leftrightarrow)$	$(\succ, \Leftarrow)$	$(\succ_d, \Leftrightarrow)$	.

Table 92: Activity relationships matrix for activities B, A, C, D ; dependency relaxation of locked dependency from A to B

XOR Split (acceptance sequences: $\{AB, AC\}$)

Locked dependency $A \Leftarrow B$

- Parallelize $\{A, B, C\}$ — **Error: Existential dependency from 'A' to 'B' is locked and cannot be changed.**